

MUHAMMAD HAMZA

Senior Software Engineer | Backend Engineering | Node.js | NestJS | TypeScript | React.js | Next.js

Phone Number: (+92) 311 435 6722	Email Address: dev.hamza.010@gmail.com
LinkedIn: https://www.linkedin.com/in/hawanh1	Address: Lahore, Pakistan
Portfolio: https://muhammad-hamza-one.vercel.app/	

ABOUT ME

Senior Software Engineer with 4+ years of experience building and maintaining production web platforms across recruitment, HR, government education, and combat sports. Backend-focused with strong experience in Node.js, NestJS, TypeScript, PostgreSQL, REST APIs, real-time systems, background jobs, and third-party integrations. Comfortable working across React, Next.js, and Vue when a product needs full-stack ownership. Experienced in schema design, performance optimization, data migrations, authentication, production support, and taking features from technical design through release.

KEY ACHIEVEMENTS

- Helped take Fightbook from first commit to production in roughly 6-7 months, leading backend technical delivery for a multi-role combat sports platform.
- Designed a flexible multi-sport data model and imported approximately 26,000 real fighter profiles with identity verification and one-click profile claiming.
- Built fight recommendations, role-based authorization, real-time chat and presence, notifications, and queue-driven background processing for Fightbook.
- Delivered Mubadala Takafo's VIP Referral backend module from schema and API design through UAT sign-off and production release.
- Optimized Justworks payroll forecasting operations from 5-10+ minutes to seconds while working with a PostgreSQL database that grew from about 60 GB to 120 GB.
- Implemented a new mathematics-related qualification flow across the UK Department for Education NPQ and ECF systems using REST APIs and background jobs.
- Migrated a legacy .NET/MySQL platform into a NestJS/PostgreSQL system, including more than 23,000 users and broader historical data.
- Built backend prototypes as the sole backend engineer using Elasticsearch for large-scale search and Kafka for high-volume social messaging workflows.

CORE COMPETENCIES

- Backend Development
- Full-Stack Development
- REST API Design & Integration
- System Architecture
- Database Design & Optimization
- Performance Optimization
- Real-Time Systems
- Data Migration & Modernization
- Authentication & Authorization
- Background Jobs & Queues
- Production Support
- Code Review & Mentoring
- Technical Delivery
- Cross-Functional Collaboration
- Debugging & Root Cause Analysis
- Task Estimation
- Technical Documentation

WORK EXPERIENCE

Senior Software Engineer

(12/2025-Present)

Aslase

Lahore, Pakistan

- Own backend-heavy full-stack features on Mubadala Takafo using NestJS, TypeScript, PostgreSQL/Prisma, Vue.js, Redis, and Azure Service Bus.
- Designed and delivered the VIP Referral module, including priority and government referral handling, candidate workflows, timed notifications, and external-partner routing.
- Built backend APIs and data models, worked through UAT to full sign-off, participated in releases, and now support production recruiter and candidate issues.
- Review junior PRs, mentor developers, estimate work, join architecture discussions, coordinate with QA, and troubleshoot production incidents.

BACKEND ENGINEER

(12/2024 - 11/2025)

Ilsa Interactive

Lahore, Pakistan

- Led backend technical delivery for Fightbook from initial schema through launch, designing NestJS/PostgreSQL workflows for six user roles, multiple sports, fights, and challenges.
- Built matchmaking recommendations, authentication, chat and presence, notifications, queues, fighter identity verification, and third-party integrations.
- Reviewed PRs, coordinated API and data contracts with frontend developers, supported production monitoring, and helped ship the first production launch in roughly 6-7 months.
- Built SPSG migration scripts that mapped legacy .NET/MySQL data into a NestJS/PostgreSQL schema with transactional validation and rejected-record tracking.

BACKEND ENGINEER

(05/2024 - 11/2024)

NeuroHeap

Lahore, Pakistan

- Worked as the sole backend engineer for FootyContact and a unified social messaging prototype, building NestJS APIs across PostgreSQL and MongoDB.
- Used Elasticsearch for fast search over athlete datasets expected to reach millions of records and Kafka to handle high-volume social interactions.
- Implemented JWT access/refresh authentication and Jest tests while owning backend architecture and API development for both products.

SOFTWARE ENGINEER / ASSOCIATE

(02/2022 - 05/2024)

SOFTWARE ENGINEER

Devsinc

Lahore, Pakistan

- Worked on Justworks payroll and forecasting optimization and the UK Department for Education NPQ/ECF services using Ruby on Rails and PostgreSQL.
- Reduced expensive payroll forecasting flows from 5-10+ minutes to seconds through query improvements, production-data corrections, and association redesign.

- Maintained and extended NPQ/ECF applications, delivered a new mathematics-related qualification flow across two repositories, and supported REST/background-job synchronization.

EDUCATION

BACHELOR OF SCIENCE IN SOFTWARE
ENGINEERING (BSSE)
PUCIT, University of the Punjab

(2018 - 2022)

Lahore, Pakistan

TECHNICAL SKILLS

- **Programming Languages:** TypeScript, JavaScript, Ruby, SQL.
- **Frontend:** React.js, Next.js, Vue.js, Redux/RTK Query, Material UI, HTML5, CSS3.
- **Backend:** Node.js, NestJS, Express.js, Hapi.js, Ruby on Rails, REST APIs, WebSockets, background jobs.
- **Databases & Data:** PostgreSQL, MongoDB, MySQL, Redis, Prisma, TypeORM, Sequelize, Elasticsearch.
- **Messaging & Async:** Kafka, BullMQ, Sidekiq, Azure Service Bus, scheduled and queue-driven processing.
- **Testing & Tools:** Jest, RSpec, Capybara, Git, GitHub, Docker, Jira, Cursor.
- **Auth & Integrations:** JWT, OAuth, Firebase Auth, Veriff, Pusher, Knock, Loops, Microsoft Teams, Odoo, Webhooks.

PROJECTS

FIGHTBOOK

- Backend-led combat sports platform supporting fighters, managers, promoters, service providers, fans, and admins across flexible sport and weight-class structures.
- Designed the schema and APIs from zero, imported about 26,000 fighter profiles, and built profile claiming, identity verification, fight/challenge workflows, recommendations, chat, notifications, and multi-role authorization.
- Stack: NestJS, Next.js, TypeScript, PostgreSQL, TypeORM, BullMQ, Redis, Firebase, Pusher, Knock, Veriff, Loops.

MUBADALA TAKAFO

- Delivered the VIP Referral module for priority and government-level recruitment workflows, including candidate fast-tracking, timed flags and notifications, external-partner routing, and interview flows.
- Built the backend APIs and data model with NestJS, PostgreSQL/Prisma, Azure Service Bus, Redis, and Vue.js, then supported UAT through full sign-off and production release.

JUSTWORKS - PAYROLL FORECASTING OPTIMIZATION

- Optimized payroll forecasting and related workflows in a large Rails/PostgreSQL system, reducing 5-10+ minute operations to seconds.
- Fixed polymorphic-association data issues and redesigned the payroll/invoice relationship while the production database grew from about 60 GB to 120 GB.

UK DEPARTMENT FOR EDUCATION - NPQ / ECF

- Maintained two Ruby on Rails applications and implemented a new mathematics-related qualification flow across registration and provider-side systems.
- Built REST synchronization, background-job support, production data fixes, automated tests, and Rails upgrade-preparation work.

SPSG LEGACY MIGRATION

- Wrote migration scripts mapping a legacy .NET/MySQL database into a new NestJS/PostgreSQL schema, including more than 23,000 users and broader historical data.
- Used transactional validation, rollback where required, and skip/error tracking for records that needed later reconciliation.

RECOMMENDATIONS

Available upon request.